

SANHITHA REDDY MANIKANTI

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SUMMARY

Master's student in Computer Science with solid foundations in software engineering, data structures, and system design. Proficient in Java, C, Python, SQL, JavaScript and familiar with Git and Linux. Hands-on experience building web applications, data pipelines and predictive models. Eager to contribute across the full software development life cycle and grow coding expertise in dynamic environments.

SKILLS AND INTERESTS

- **Programming:** Python, Java, C, SQL, JavaScript
- **Tools & Frameworks:** Git, Linux, HTML, CSS, JavaScript, Power BI, Java Spring Framework
- **Core Areas:** Software Engineering, Data Structures, Databases, System Design

INTERNSHIP EXPERIENCE

Research Assistant University at Buffalo	Dec 2024 – Dec 2025
• Developed machine learning pipelines for handwriting analysis, integrating computer vision and NLP to extract features from 500+ handwriting samples, identifying language and writing challenges. • Implemented and benchmarked supervised models including Logistic Regression, Random Forest, XGBoost, and CNN architectures using Python and scikit-learn, improving predictive accuracy by 15–20% over baseline. • Conducted cross-validation and evaluation with accuracy, precision, recall, and F1-score metrics, delivering reproducible results and actionable insights for research documentation.	
Code Clause, Hyderabad, India	Sep 2023 – Dec 2023
Data Science Intern	Jun 2025 – Aug 2025
• Preprocessed and analyzed 50,000+ records using Python and SQL, improving data quality and consistency. • Built and optimized predictive models, achieving 10–15 improvement in accuracy over baseline, and delivered actionable insights to stakeholders. • Created interactive Python dashboards to visualize key metrics and trends, and managed Git-based workflows to ensure reproducible pipelines and version control for project teams.	

EDUCATION

University at Buffalo, SUNY	Aug 2024 – Dec 2025
Masters in Computer Science and Engineering (AI/ML Track)	GPA: 3.5/4
Mahatma Gandhi Institute of Technology	2020 – 2024
Bachelor of Engineering, Computer Science and Engineering	GPA: 3.3/4

ACADEMIC PROJECTS

Skills Gap Analysis with Generative AI — Python, FastAPI, React, scikit-learn, spaCy, OpenAI API, MongoDB, Docker	
• Developed an AI system to analyze 10,000+ resumes and job descriptions, identify skill gaps, and generate personalized upskilling paths. • Leveraged spaCy NER, embedding-based semantic matching, and LLMs to generate actionable career insights and missing-skill recommendations. • Built dual dashboards for job seekers and educators, aligning skills and curriculum with real-time market demand. • Implemented job-market scraping, skill clustering, and analytics, enabling data-driven workforce planning and reporting.	
End-to-End Cricket Analytics Pipeline — Snowflake, SQL, JSON, CDC, Snowsight	
• Architected a multi-layer Snowflake data warehouse (Landing, Raw, Clean, Consumption) to ingest and transform deeply nested JSON cricket data into structured formats. • Engineered automated ETL pipelines using Snowflake Streams, Tasks, and SQL to implement Change Data Capture (CDC) and continuous data orchestration. • Designed a Star Schema with optimized Fact and Dimension tables, enabling high-performance analytics and visualizations for cricket insights. • Reduced manual data processing time by 70	

CERTIFICATIONS

- CPA: Programming Essentials in C++ and Networking Essentials from Cisco
- The Joy of Computing using Python
- Java Foundations Certificate from Oracle and Database Programming with SQL from Oracle